Application of Three-Dimensional Interactive Virtual Conference in Special Education

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Abstract. In special education, the social skills and learning needs of deaf and mute students have always been a focus of attention. With the advancement of educational reforms, many learning tasks and communication activities are conducted through conference systems. However, traditional 2D conference systems are not friendly to special students, as their physiological challenges often lead to feelings of inferiority in communication, affecting participation effectiveness. To address this issue, we propose a Three-Dimensional Interactive Virtual Conference System (3DIVCS), which not only provides video, voice communication, and speech recognition features, but also enables users to interact and engage on stage as 3D virtual characters. To validate the effectiveness of 3DIVCS for deaf and mute individuals, researchers developed a set of 3DIVCS applications and conducted real-world experiences, surveys, interviews, and analyses using independent t-tests with 20 deaf and mute students. The results indicate that 3DIVCS significantly improved their learning outcomes in conferences, enhanced their confidence, and ultimately elevated their sense of participation and communication quality.

Keywords: Three-Dimensional Interactive Virtual Conference System · Special Education · Deaf and Mute Students

1 Introduction

In the field of special education, the social skills and learning needs of deaf and mute students have always been a focal point for educators. With the continuous advancement of educational reforms, an increasing number of learning tasks and communication activities are being conducted through online conferencing systems [1]. However, traditional two-dimensional conferencing systems are not user-friendly for special groups, particularly for deaf and mute students. Due to their physiological challenges, these students often feel inferior when communicating, which negatively impacts their participation and learning efficiency [2]. How to enhance their learning engagement and social confidence in virtual environments has become an urgent issue in the current process of digitalizing special education.

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For many years, conferencing systems have been used to address the issue of not being able to hold meetings in the same physical space. Many work meetings, academic conferences, or online courses often use conferencing systems as communication tools. The development of internet technology has accelerated the adoption of conferencing systems, making them a tool for communication among students, employees, and friends. Some scholars conducted a survey on students' willingness to continue using conferencing systems for course learning after the pandemic, and the results showed that even after the pandemic, users still expressed a positive attitude towards the use of virtual conferencing systems [3].

According to the World Federation of the Deaf (WFD), more than 5% of the world's population (about 430 million people) suffer from disabling hearing loss, including 432 million adults and 34 million children [4]. Although some scholars are dedicated to sign language research, communication difficulties arise due to the fact that most people have trouble understanding the semantic information of sign language [5]. Deaf-mutes are prone to communication and emotional problems, making them prone to become introverted and resistant to social and face-to-face socializing. Because deaf-mutes cannot communicate with ordinary people normally, they also suffer from low self-esteem [6]. As a special group of people, deaf-mutes have the same needs of study, work and communication as normal people. As a result of the influence of physical defects, the social ability and expression ability differ from the ordinary people.

In the field of special education, the social skills and learning needs of deaf and mute students have always been a major concern for educators. With the ongoing advancement of educational reforms, an increasing number of learning tasks and communication activities are conducted through online conferencing systems. However, traditional two-dimensional conferencing systems are not user-friendly for special groups, especially deaf and mute students. Due to their physiological challenges, these students often feel inferior when participating in communication, which negatively impacts their participation and learning efficiency [7]. Enhancing these students' engagement and social confidence in virtual environments has become an urgent issue in the current process of digitalizing special education.

Traditional two-dimensional conferencing systems primarily rely on screen sharing, camera footage, and audio transmission, which may be sufficient for regular users. However, for the deaf and mute community, merely relying on visual elements and limited text information falls far short of meeting their communication needs [8]. Since they are unable to interact with others through speech, deaf and mute students often remain passive observers in these conferencing systems, lacking a sense of participation and interactivity [9]. This not only reduces their engagement in learning but also exacerbates their feelings of inferiority and social barriers.

In the 2D conference system, deaf-mute users can participate in the meeting as ordinary users, but in many cases, it is difficult for all participants to interact effectively and they can only act as an audience. With the development of 3D technology, 3D technology is also applied to virtual conference system. After the meeting system is transformed from 2D space to 3D space, users can exist as virtual characters in 3D space. Users can interact with the scene and view different angles of the 3D space freely. When the meeting is at rest, each virtual role can roam and interact in the 3D space, which enhances the

immersive effect of the meeting and effectively alleviates users' participation fatigue. Some scholars used the virtual world as a conference environment to enable users to have immersive meetings and conduct high-quality media interaction, and recorded user experience by means of questionnaires and interviews. The experimental results show that participants tend to cooperate and communicate in the three-dimensional virtual environment [8]. In the past research of conference system, many scholars pay attention to the research of network communication and function of conference system. On the research of whether virtual conference system is suitable for deaf-mutes, few scholars have put forward effective research results. In order to enable the deaf-mutes to communicate with ordinary people, experts have opened intelligent apps to help them communicate with each other. Alkadhi et al. [10] developed mobile applications for deaf-mums to help them communicate with dentists. In the survey, it was also found that deaf-mums were not excluded from communicating with others with the help of smart phones, tablets and computers, and could achieve good communication effects.

In order to improve the teaching effect and learning enthusiasm of special students, virtual reality (VR) and three-dimensional interactive technologies offer us innovative solutions. The introduction of the Three-Dimensional Interactive Virtual Conference System (3DIVCS) not only enables users to communicate via video and audio but also enables them to interact as three-dimensional virtual characters. In this virtual environment, users can move freely, choose different perspectives to observe and interact, greatly enhancing the immersion and interactivity of the conference. For deaf and mute students, the representation of these three-dimensional virtual characters can conceal their physical limitations in real life, helping them engage in social interactions in an equal virtual environment, thus boosting their confidence and sense of participation. The design of the 3DIVCS system not only includes basic conference functions such as video communication, voice dialogue, screen sharing, and speech recognition, but also enables users to interact within virtual scenes through virtual characters. This approach not only breaks the passive limitations of users in traditional two-dimensional conferencing systems but also enhances users' focus and engagement through immersive experiences. Moreover, the speech recognition function provides real-time captions for deaf and mute students, helping them better understand the conference content and actively participate in discussions.

2 Related Work

In recent years, the rapid development of computer networks and virtual reality technology has led to significant advancements in virtual conference systems. These systems use virtual reality technology to combine images, sounds, and text, allowing people to communicate face-to-face regardless of their physical location. Virtual conference systems have been widely used to address meeting, learning, and communication problems, as well as to promote physical activity and health campaigns. Research has shown that digital platforms can be key in supporting physical activity participation, especially during stay-at-home restrictions such as those imposed during the COVID-19 pandemic [11]. Numerous research institutions and enterprises both domestically and abroad have conducted in-depth research on virtual conference systems. Microsoft's MSN and Tencent's QQ offer group chat with a 2D interface, while Tencent and Zoom have developed

2D virtual conference systems that allow for screen and camera sharing, as well as voice transmission. Researchers have focused on a range of communication technology issues in virtual conference systems, with many studying network communication and conference information security. For example, Isobe et al. [12] analyzed and evaluated encryption protocols in virtual conference systems to protect dialogue information, while C. Jang et al. [13] proposed a distributed conference operating system structure to relieve network loads for multiple users in virtual conference systems.

Several researchers have investigated the participation and attentiveness of users in virtual conference systems. For example, Erickson et al. conducted a three-day conference in a virtual environment and evaluated different forms of virtual conferences, leading to discussions about their potential [14]. The conventional 2D virtual conference system often fails to keep users focused during usage. To address this issue, Kamel et al. [15] developed a 2D virtual conference system that tracks users' engagement and enables them to indicate their involvement in the meeting with a single virtual swipe. Moreover, researchers have explored various ways to enhance the social communication of deaf individuals [16, 17]. Savvas et al. [18] studied an interactive tool that leverages computer vision, speech recognition, and tactile sign language analysis to aid deaf users in using computers for gaming, thereby enhancing their social interaction. Similarly, Yousaf K et al. [19] developed a mobile application that improves communication for both deaf individuals and the general population, which uses Meir frequency cepstral coefficients for feature extraction, a hidden Markov model toolkit for speech recognition, and 3D virtual characters for visualization. This intervention has helped to enhance face-to-face social interactions for deaf individuals with the aid of mobile technology.

Creative teams are increasingly developing portable VR glasses for 3D virtual space, with the proposed display system offering advantages such as ease of use, low production cost, high portability and mobility. This technology is expected to be widely used in the future for 3D virtual space [20]. Additionally, Ruofei Du et al. [21] have constructed an interactive mixed reality social media platform that includes a geotagged virtual world for users to chat and collaborate. The study found that the platform has strong interactivity and creativity, which is conducive to the development of social activities. As such, it is anticipated that users will eventually move from traditional text and 2D media to 3D media. Ibanez et al. [22] suggest that while the learning process of the traditional learning management system has been widely studied, research literature on user learning in 3D virtual worlds is relatively limited. However, the 3D virtual world is a promising environment that enables users to immerse themselves as virtual characters and improve their concentration.

The social abilities of deaf-mutes can be severely impacted by their introverted nature in social situations and their difficulty using traditional communication methods like voice-based virtual conference software. To address this issue, researchers have explored the application of gesture recognition technology to convert gestures into words and assist with communication [19]. Another approach involves using mobile applications to convert spoken language into sign language through 3D virtual characters, as suggested by J. P. Bigham et al. [23]. However, converting spoken language into sign language can result in information omission and hinder understanding for deaf-mutes who are already able to recognize written language. To improve the user experience and enthusiasm of

deaf-mutes in virtual conference systems, this study proposes a 3D interactive virtual conference system based on the 2D virtual conference system. This new system aims to facilitate smoother communication and interaction for deaf-mutes in virtual conference environments.

3 Proposed Methodology

3.1 Enhancing Functionality of the 3D Interactive Virtual Conference System

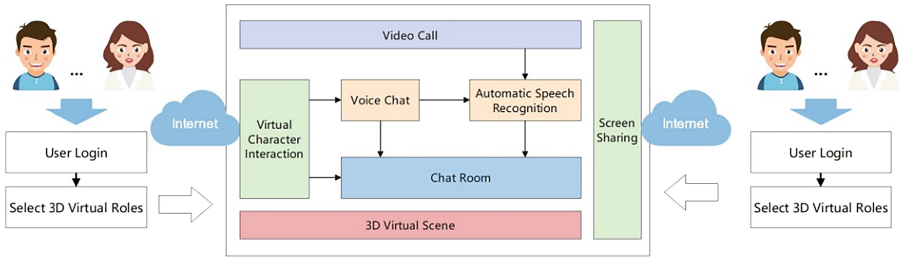


Fig. 1. Function module structure of 3D interactive virtual conference system.

This study developed an interactive virtual conferencing system based on a three-dimensional virtual environment, with the system architecture illustrated in Fig. 1. Unity3D 2022.2.1 was employed as the development engine, and coding was conducted in Visual Studio 2019, with MySQL utilized for data storage. Three-dimensional scenes and character models were edited and UVW-mapped in 3ds Max 2019, after which the models were imported into the Unity engine for further texture mapping and animation development. Functionalities were implemented within Unity, encompassing a user login module, 3D virtual character selection module, conference room connection module, 3D scene module, screen sharing, video calling, voice communication, speech-to-text module, virtual character interaction and dialogue module (e.g., exchanging business cards), and a text communication module.

Figure 1 illustrates the process of using the 3D Interactive Virtual Conference System (3DIVCS). Firstly, users are required to log in to the system with their username and select their 3D virtual character model. This character will represent the user in the system and participate in the meeting. Once logged in, users are assigned a seat in the 3D virtual meeting scene. During the meeting, the user's screen displays the 3D virtual environment, which can be freely navigated during breaks. At the beginning of the meeting, users are allocated a seat in the virtual conference room, which consists of three screens. The main screen is used for screen sharing, while the two secondary screens display camera images and subtitles generated from voice-to-text conversion. In particular, the secondary screens serve as an important tool for deaf-mute users who cannot hear the audio content. Figure 2 demonstrates the use of the secondary screen for accessing conference content. During breaks and after the meeting, users are free to roam around the 3D virtual environment and interact with other 3D virtual characters

through collision detection. Users can also exchange personal business cards via text or voice, subject to the consent of the other party.

3.2 Enhancing User Participation in Virtual Conferences Through the 3D Interactive Virtual Conference System



Fig. 2. An user perspective interface of 3D virtual conference system.

The effect of user participation in virtual conference systems has been a topic of interest for many researchers. The 3D Interactive Virtual Conference System (3DIVCS) incorporates three screens, with the main screen used for displaying users' PPTs or screen sharing, while the other two display camera footage and subtitles generated by speech recognition. By placing speech recognition as a primary feature in the scene, the 3DIVCS enables deaf-mute users to access conference communication content via the subtitle screen. This feature is highly beneficial for deaf-mute users, as it enables them to see the conference screen and follow the speech content. Moreover, during long meetings, users can adjust their perspective to view other users around them, which facilitates alleviate the fatigue associated with prolonged viewing of a static image.

3.3 The Benefits of Using 3D Virtual Characters in the 3DIVCS for Deaf-Mute Users

Incorporating 3D virtual characters as users in the 3D Interactive Virtual Conference System (3DIVCS) provides several advantages, especially for deaf-mute users. Research by Halbreich et al. [24] suggests that deaf-mutes often experience communication barriers due to congenital reasons, which can result in a lack of confidence and inferiority, and severe communication problems with others. Using 3D virtual characters equalizes all users and facilitates deaf-mute users conceal their limitations. In the 3DIVCS, users are



Fig. 3. A dialog interface between two user avatars in a 3d virtual conference system.

randomly assigned seats in the 3D virtual scene. Before the meeting starts or during a break, users can move around freely and initiate conversations with other virtual characters through collision detection, as shown in Fig. 3. When two characters are close, they can exchange personal business cards and chat applications, which can be followed by video, voice or text conversations. Using virtual characters can help deaf-mute users feel more expressive and confident, enabling them to better participate in the conference.

4 Experimental Results and Discussions

4.1 Participants

This study examines the efficacy of 3DIVCS for deaf-mute users. The research randomly selected 20 deaf-mute students (10 in grade 2 and 10 in grade 3) from the Special Education College of Fuzhou Vocational and Technical College of China. Participants were informed that they could withdraw from the study at any time if they felt uncomfortable. The students were randomly divided into two groups, each with the same prior knowledge. Two virtual conference systems were used in the experiment: the TENCENT Conference system, a 2D system without 3D virtual scenes or roles, and the 3D Interactive Virtual Conference System (3DIVCS) developed by the research team. The 3DIVCS incorporates 3D virtual roles and voice-to-text functionality, whereas the TENCENT Conference system is moderated and referred to as 2DVMS.

4.2 Experimental Procedure

The main focus of this study was to investigate the effectiveness of a 70-min virtual meeting, moderated by a teacher, in facilitating learning and communication related to professional knowledge. The meeting included a 10-min break every 30 min. The meeting was presided over by the teacher. The first group conducted the experiment in

a 2D virtual conference system, the first group was the reference group and the second group was the experimental group, and the experiment was conducted in a 3D interactive virtual conference system. The members of Group one and group two are all using the conference system in different places. In order to better understand the participation effect of deaf-mute users using 3D interactive virtual conference system, we adopted the five-point quantification table of cognitive load proposed by Hwang et al. [25] for development. This post-test usefulness sets The questionnaire in terms of Perceived usefulness and Perceived ease of use of The virtual meeting system. Among them, the perceived usefulness evaluation of virtual reality conference system has 6 questions in the questionnaire, and the perceived ease of use evaluation of virtual reality conference system has 7 questions in the questionnaire. The maximum score for each answer is 5 points, with a minimum of 1 point.

Each group was led by a professional teacher who spent 15 min at the start of the experiment introducing the main features of the virtual conference system. During the 60-min professional learning activity, each group used either the 2D virtual conference system (group one) or the 3D interactive virtual conference system (group two) and then filled out usability questionnaires at the end of the meeting. During the break, the participants in group two were encouraged to interact with each other using avatars. The primary differences between the two groups were the virtual meeting systems they used and the methods they used to communicate with other group members. Following the experiment, the moderator conducted interviews with and summarized the experiences of the two groups of deaf students.

4.3 Results

After conducting the questionnaire survey, we utilized an independent t-test to analyze the data samples. An independent sample t-test is suitable for testing the difference in data obtained from two groups of unrelated samples. In this study, we used two different groups to conduct experiments on the 2D virtual conference system and 3D virtual conference system, respectively. As these groups met the conditions for the use of independent t-test, we adopted this method for statistical analysis. Table 1 presents the experimental results, with the mean representing the average value of the experimental data and SD reflecting the degree of dispersion among individuals in the group. The t-test was utilized to compare the differences between the two conference systems, analyzing the relationship between categorical and quantitative data.

An independent sample t-test was conducted to compare the post-test results of the two groups. The results showed a significant difference ($t = 5.635$, $P < 0.05$), indicating that the 3D interactive virtual conference system effectively facilitates communication for deaf-mute users. However, in terms of human-computer interaction, the post-test did not reach a significant level ($t = 0.000$, $P > 0.05$), indicating that the 3DIVCS and 2D virtual conference system have similar ease of human-computer interaction. Regarding the perceived usefulness, the post-test of the two groups was significantly different ($t = 7.019$, $P < 0.05$), with a mean value of 4.53, indicating that the 3DIVCS was more useful than the 2D virtual conference system. However, in terms of perceived ease of use, the post-test results did not show a significant difference ($t = 0.397$, $P > 0.05$), suggesting that 3DIVCS and 2D virtual conference system have similar ease of use, and

Table 1. Independent sample t-test the congnitive loads between the two groups.

	Group	*N*	*Mean*	*SD*	*t*
Help with communication	3DIVCM	10	4.70	0.483	5.635**
	2DVCM	10	2.60	1.07	
Human Computer Interaction	3DIVCM	10	4.90	0.316	0.000
	2DVCM	10	4.90	0.316	
Perceived usefulness	3DIVCM	10	4.53	0.204	7.019**
	2DVCM	10	3.48	0.426	
Perceived ease of use	3DIVCM	10	4.80	0.204	0.397
	2DVCM	10	4.77	0.099	
Total	3DIVCM	10	4.67	0.113	7.494**
	2DVCM	10	4.17	0.177	

** $p < 0.05$

deaf-mute users do not encounter operational difficulties. Overall, the results indicate that 3DIVCS is beneficial for deaf-mutes to communicate with other users online, and the perceived usefulness of the 3DIVCS system is higher than that of the 2D virtual conference system. In terms of human-computer interaction and perceived ease of use, 3DIVCS is similar to 2D virtual conference system, which is relatively easy to operate.

4.4 Interview

An additional interview section was included in the experiment, following the interview method developed by Hwang et al.[26]. The interview consisted of four questions designed to assess the impact of the 3D interactive virtual conferencing system on deaf-mute users. The interview questions included: "What are the differences between the 3D interactive virtual conferencing system and the 2D conferencing system you have used before?", "Did this meeting system help you to participate in online meetings? If so, how?", "Overall, what are the advantages of the 3DIVCM system?", and "What kind of support did you receive through this system?" To gain a comprehensive understanding of the impact of the 3D interactive virtual conferencing system on deaf-mute students, five participants were randomly selected from each group to participate in the interview, resulting in a total of 10 participants. The comparison of the frequency of mentions for 3DIVCM and 2DVCM across interviews on different topics and content is shown in Table 2.

During the interview, U01 from the 3DIVCM group expressed that the communication mode of 3D characters is similar to that of games, which makes communication feel more relaxed and enjoyable. U02 appreciated the simplicity of the 3DIVCM operation, which allowed him to communicate with ease and without pressure. In contrast, U06 from the 2DVCM group did not feel pressured to interact, but also had limited contact and communication with others in the system. U09 mentioned that the 2DVCM display screen being controlled by the host can be tiring over time. Overall, the 3DIVCM appears

to be more helpful in facilitating communication and providing a more relaxed communication experience for deaf-mute users. Additionally, the 3DIVCM system shows better man-machine operation performance and enables deaf-mutes to participate more actively in meetings. The operation difficulty of the 3DIVCM system did not pose any significant challenges for deaf-mute users, who found the system to be easy to use.

Table 2. The analysis results of the interviews.

Theme	Main content	The number of times mentioned	
		3DIVCM	2DVCM
Improve frequency of communication	Initiate communication with others	2	1
	Relaxing	2	1
	Ask for Help	2	0
Improve motivation to attend virtual meetings	I'll spend more time operating the conference system	2	0
	Always focus on the conference system	3	0
	Love the meeting format	1	1
The operation difficulty of the system	Can pick it up quickly	3	3
	Not hard	3	3
	Receivability	3	3
Acceptence	Better than the past Conference system	4	1
	Hope taking all Conference will be this	3	2
	Try again in the future	2	2

3DIVCM and *2DVCM* represent users of different systems, respectively.

5 Conclusion

With the rapid development of internet technology, online meeting systems and online classrooms have gradually been recommended for special education reform efforts. However, traditional online meeting systems have not considered the needs of deaf-mute users, resulting in a poor participation experience for them. To address this issue, we developed a 3D Interactive Virtual Conference System (3DIVCS) that takes into account the learning styles and individual characteristics of deaf-mute users. In addition to providing multi-user video communication, voice communication, and speech recognition functionalities, the system enables users to communicate and interact within a virtual scene through 3D avatars. To evaluate the effectiveness of the 3D Interactive Virtual

Conference System (3DIVCS), we conducted a user experience experiment with deaf-mute participants and compared it with a 2D Virtual Conference System (2DVCS) using questionnaires and interviews. Our findings show that 3DIVCS outperformed 2DVCS in terms of "facilitating communication" and "perceived usefulness," while there was no significant difference in human-computer interaction and perceived ease of use. During the interview phase, 3DIVCS users reported feeling more motivated to communicate and interact when using virtual avatars, which is consistent with previous research on gamification and learning motivation. Participants also found no significant difference between 3DIVCS and 2DVCS in terms of system operation difficulty and ease of use. We believe that the 3D Interactive Virtual Conference System can significantly improve the participation experience for deaf-mute users and could bring benefits to all users in the future. We plan to conduct more experimental tests with other user groups. In the future, we will expand the research group and conduct further research work on the social interaction of deaf-mute people with the aid of metaverse technology.

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Data Availability. We provided the information about the research purpose, possible risk, discomfort, human rights, and choices during the research implementation to the teachers and students who voluntarily participated in this research. They knew that their participation was voluntary and they could withdraw from the study at any time. The data can be obtained by sending a request e-mail to the corresponding author.

Disclosure of Interest. The author declares that there are no conflicts of interest.

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